

ITT-216- RIP and OSPF Single Area Configuration

Professor: Tim Collins

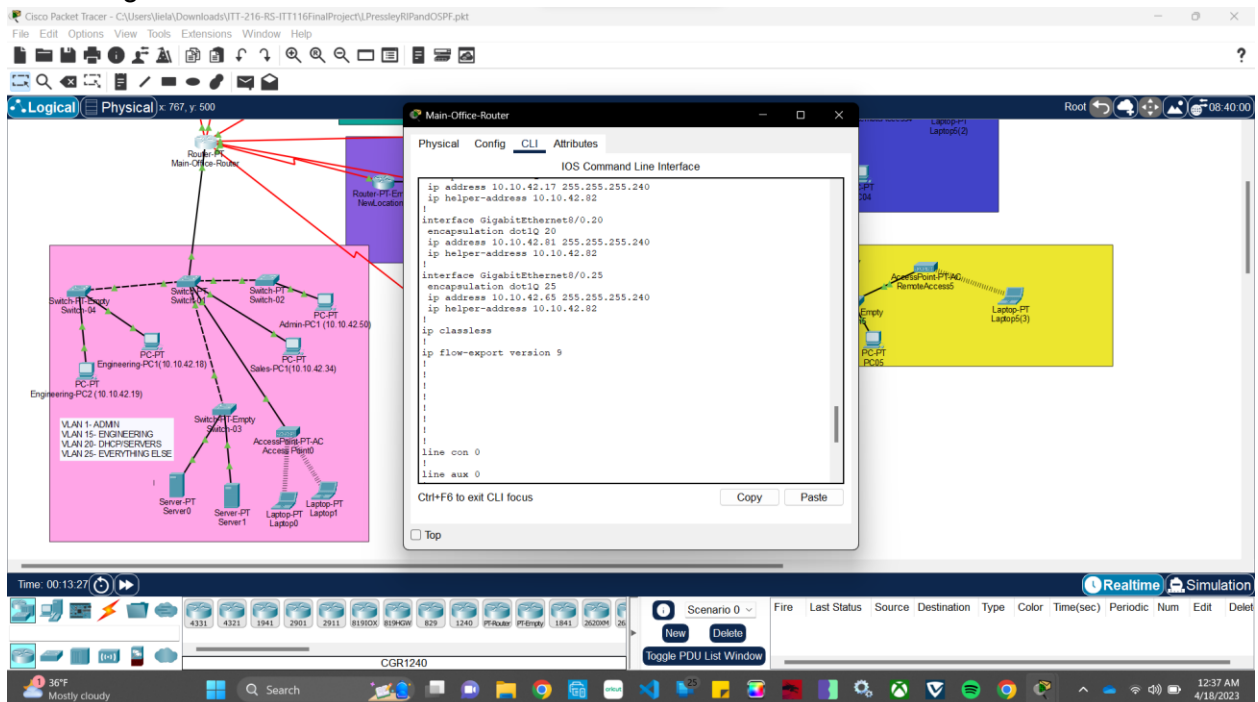
Course: ITT-216 Enterprise Route & Switch

Student: Liela Pressley

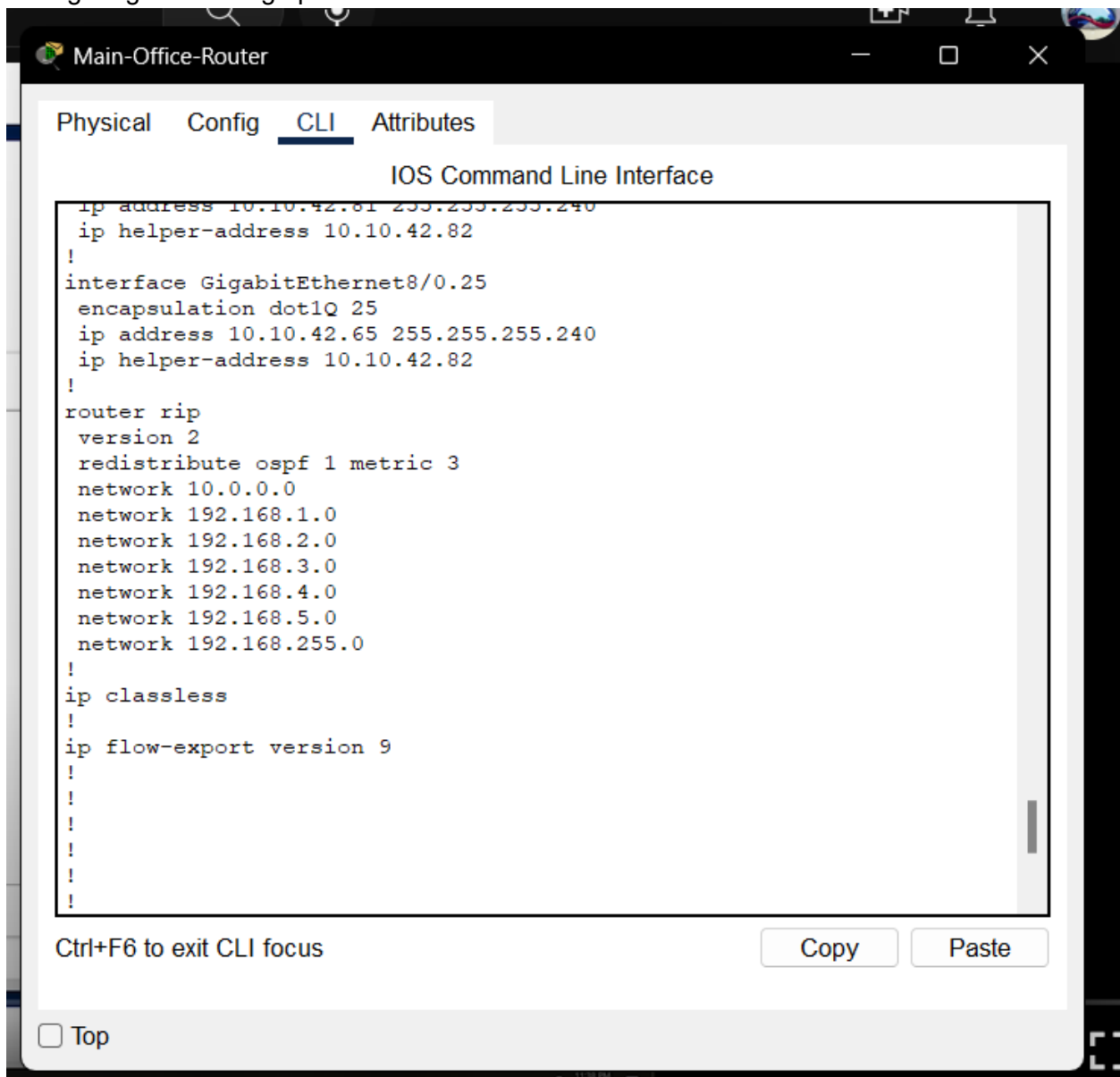
Date: 4/16/2023

Title: RIP and OSPF Single Area Configuration

Removing static routes



Configuring and setting up RIP and OSPF



Main-Office-Router

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
network 192.168.1.0
network 192.168.2.0
network 192.168.3.0
network 192.168.4.0
network 192.168.5.0
network 192.168.255.0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
end

Main-Office-Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Main-Office-Router(config)#router ospf 1
Main-Office-Router(config-router)#log-adjacent
                                     ^
% Invalid input detected at '^' marker.

Main-Office-Router(config-router)#log-adjacency-changes
Main-Office-Router(config-router)#redistribute rip metric 50000
subnets
Main-Office-Router(config-router)#192.168.255.0 0.0.0.3 area 0
                                     ^
% Invalid input detected at '^' marker.

Main-Office-Router(config-router)#network 192.168.255.0 0.0.0.3 area
0
Main-Office-Router(config-router)#
```

Ctrl+F6 to exit CLI focus

CopyPaste

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12:5

IP a Main-Office-Router

Physical Config CLI Attributes

IOS Command Line Interface

```
!
line vty 0 4
  login
!
!
!
end

Main-Office-Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Main-Office-Router(config)#router ospf 1
Main-Office-Router(config-router)#log-adjacent
^
% Invalid input detected at '^' marker.

Main-Office-Router(config-router)#log-adjacency-changes
Main-Office-Router(config-router)#redistribute rip metric 50000
subnets
Main-Office-Router(config-router)#192.168.255.0 0.0.0.3 area 0
^
% Invalid input detected at '^' marker.

Main-Office-Router(config-router)#network 192.168.255.0 0.0.0.3 area
0
Main-Office-Router(config-router)#network 192.168.255.4 0.0.0.3 area
0
Main-Office-Router(config-router)#network 10.10.42.16 0.0.0.15 area 0
Main-Office-Router(config-router)#network 10.10.42.32 0.0.0.15 area 0
Main-Office-Router(config-router)#network 10.10.42.48 0.0.0.15 area 0
Main-Office-Router(config-router)#network 10.10.42.64 0.0.0.15 area 0
Main-Office-Router(config-router)#network 10.10.42.80 0.0.0.15 area 0
Main-Office-Router(config-router)#network 10.10.42.96 0.0.0.15 area 0
Main-Office-Router(config-router)#network 10.10.42.112 0.0.0.15 area
0
Main-Office-Router(config-router)#network 10.10.42.120 0.0.0.15 area
0
Main-Office-Router(config-router)#network 10.10.42.144 0.0.0.15 area
0
Main-Office-Router(config-router)#exit
Main-Office-Router(config)#exit
Main-Office-Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Main-Office-Router#
```

Ctrl+F6 to exit CLI focus

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☐ Top

12:55 AM
4/18/2023

Physical Config CLI Attributes

IOS Command Line Interface

```
Remote-Office-1#
Remote-Office-1#
Remote-Office-1#
Remote-Office-1#
Remote-Office-1#
Remote-Office-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Remote-Office-1(config)#no ip route 10.10.42.0 255.255.255.0
Serial0/0
Remote-Office-1(config)#router ospf 1
Remote-Office-1(config-router)#log
Remote-Office-1(config-router)#log-adjacency-changes
Remote-Office-1(config-router)#network 10.10.42.128 0.0.0.15 area 0
Remote-Office-1(config-router)#network 10.10.42.144 0.0.0.15 area 0
Remote-Office-1(config-router)#network 10.10.42.112 0.0.0.15 area 0
Remote-Office-1(config-router)#
Remote-Office-1(config-router)#exot
      ^
% Invalid input detected at '^' marker.

Remote-Office-1(config-router)#exit
Remote-Office-1(config)#exit
Remote-Office-1#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Remote-Office-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

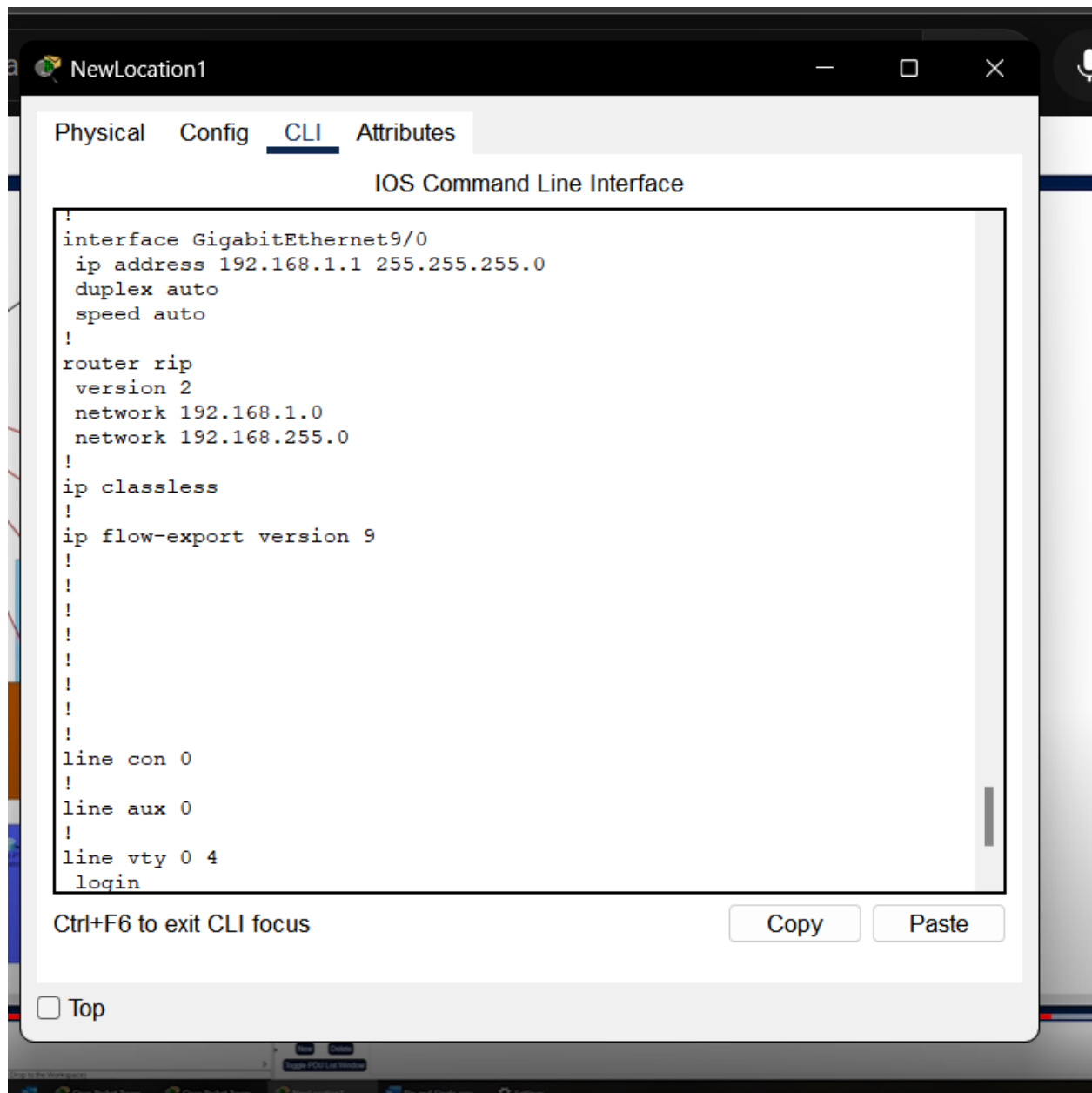
    10.0.0.0/28 is subnetted, 2 subnets
C       10.10.42.96 is directly connected, GigabitEthernet9/0.30
C       10.10.42.112 is directly connected, GigabitEthernet9/0.35
    192.168.255.0/30 is subnetted, 1 subnets
C       192.168.255.0 is directly connected, Serial0/0

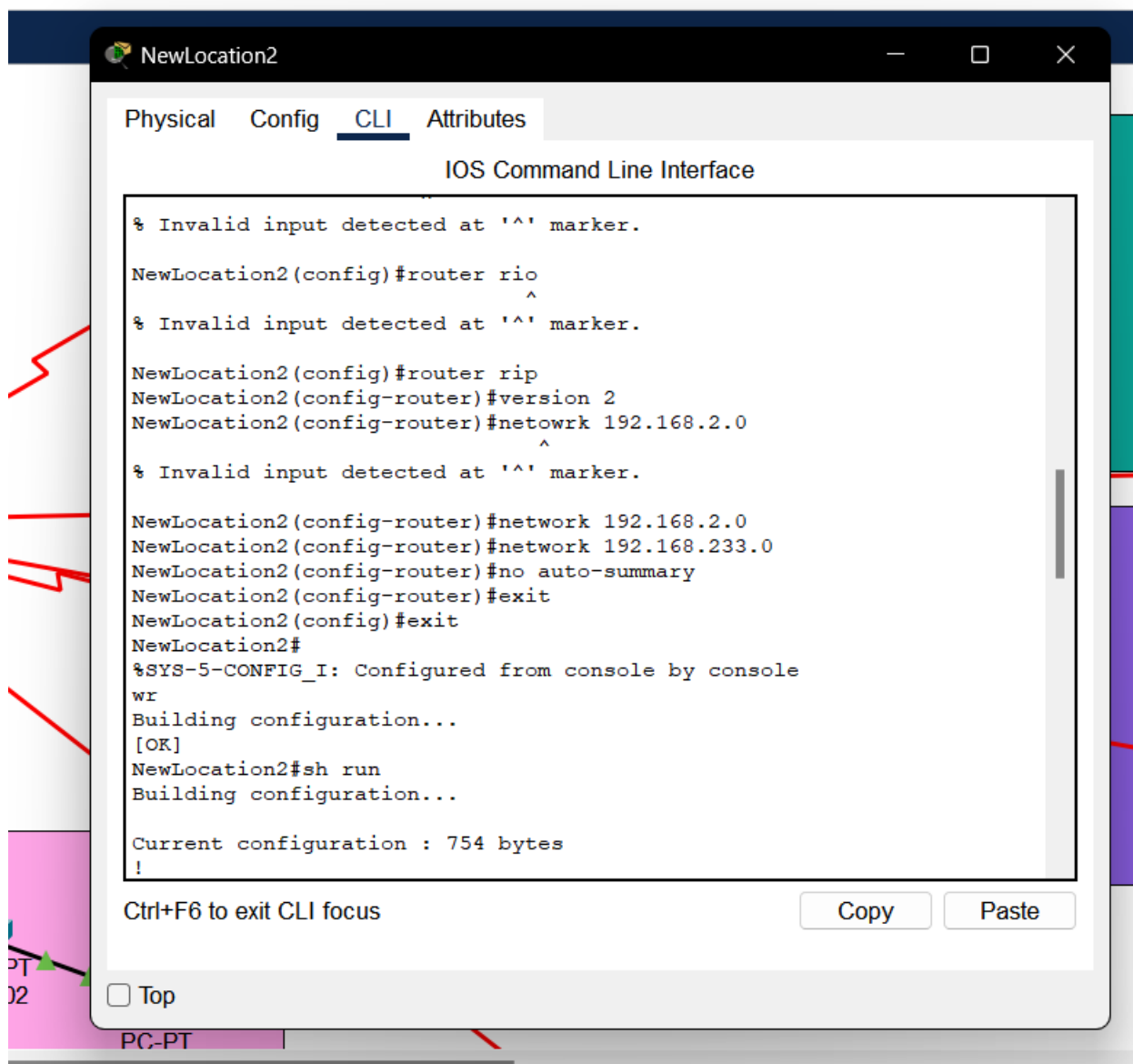
Remote-Office-1#
```

Ctrl+F6 to exit CLI focus

Copy

Paste





Cisco Packet Tracer - C:\Users\jlela\Downloads\ITT-216-RS-ITT116FinalProject\PresleyRIPandOSPF.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x:780, y:601

NewLocation2

Physical Config CLI Attributes

IOS Command Line Interface

```
interface GigabitEthernet8/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet9/0
ip address 192.168.2.1 255.255.255.0
duplex auto
speed auto
!
router rip
version 2
network 192.168.2.0
network 192.168.233.0
no auto-summary
!
ip classless
!
ip flow-export version 9
!
```

Ctrl+F6 to exit CLI focus

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Top

Time: 01:06:54

819HGW

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

New Delete

Toggle PDU List Window

57°F Sunny

6:16 PM 4/18/2023

Cisco Packet Tracer - C:\Users\jlela\Downloads\ITT-216-RS-ITT116FinalProject\PresleyRIPandOSPF.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x:1014, y:979

NewLocation3

Physical Config CLI Attributes

IOS Command Line Interface

```
interface GigabitEthernet8/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet9/0
ip address 192.168.3.1 255.255.255.0
duplex auto
speed auto
!
router rip
version 2
network 192.168.3.0
network 192.168.255.0
no auto-summary
!
ip classless
!
ip flow-export version 9
!
```

Ctrl+F6 to exit CLI focus

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More--

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VLAN 1-ADMIN
VLAN 15-ENGINEERING
VLAN 20-DHCP-SERVERS
VLAN 25-EVERYTHING ELSE

Time: 01:07:37

819HGW

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

New Delete

Toggle PDU List Window

57°F Sunny

6:17 PM 4/18/2023

Cisco Packet Tracer - C:\Users\jlela\Downloads\ITT-216-RS-ITT116FinalProject\PresleyRIPandOSPF.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1325, y: 527

NewLocation4

Physical Config CLI Attributes

IOS Command Line Interface

```
duplex auto
speed auto
interface GigabitEthernet9/0
ip address 192.168.4.1 255.255.255.0
duplex auto
speed auto
router rip
version 2
network 192.168.4.0
network 192.168.255.0
no auto-summary
ip classless
ip flow-export version 9
line con 0
--More--
```

Ctrl+F6 to exit CLI focus

Copy Paste

Time: 01:05:35

819HGW

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

New Delete

Toggle PDU List Window

57°F Sunny

6:14 PM 4/18/2023

Cisco Packet Tracer - C:\Users\jlela\Downloads\ITT-216-RS-ITT116FinalProject\PresleyRIPandOSPF.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1169, y: 971

NewLocation5

Physical Config CLI Attributes

IOS Command Line Interface

```
interface GigabitEthernet8/0
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet9/0
ip address 192.168.5.1 255.255.255.0
duplex auto
speed auto
router rip
version 2
network 192.168.5.0
network 192.168.255.0
no auto-summary
ip classless
ip flow-export version 9
line con 0
--More--
```

Ctrl+F6 to exit CLI focus

Copy Paste

Time: 01:04:57

819HGW

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

New Delete

Toggle PDU List Window

57°F Sunny

6:13 PM 4/18/2023

The screenshot shows a window titled "Main-Office-Router" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface".

The CLI output shows the following configuration and status:

```

R1 - OSPF NSSA External type 1, R2 - OSPF NSSA External type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/28 is subnetted, 7 subnets
C    10.10.42.16 is directly connected, GigabitEthernet8/0.15
C    10.10.42.32 is directly connected, GigabitEthernet8/0.10
C    10.10.42.48 is directly connected, GigabitEthernet8/0.1
C    10.10.42.64 is directly connected, GigabitEthernet8/0.25
C    10.10.42.80 is directly connected, GigabitEthernet8/0.20
O    10.10.42.128 [110/65] via 192.168.255.5, 00:16:52, Serial1/0
O    10.10.42.144 [110/65] via 192.168.255.5, 00:16:52, Serial1/0
192.168.255.0/30 is subnetted, 7 subnets
C    192.168.255.0 is directly connected, Serial0/0
C    192.168.255.4 is directly connected, Serial1/0
C    192.168.255.8 is directly connected, Serial2/0
C    192.168.255.12 is directly connected, Serial3/0
C    192.168.255.16 is directly connected, Serial4/0
C    192.168.255.20 is directly connected, Serial5/0
C    192.168.255.24 is directly connected, Serial6/0

Main-Office-Router>
Main-Office-Router>
  
```

At the bottom of the CLI window, there is a prompt "Ctrl+F6 to exit CLI focus" and buttons for "Copy" and "Paste".

Cisco Packet Tracer - C:\Users\leila\Downloads\ITT-216-RS-ITT116FinalProject\PresleyRIPandOSPF.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 589, y: 636

Root 02:22:30

Main-Office-Router

Physical Config CLI Attributes

IOS Command Line Interface

```
Main-Office-Router>
Main-Office-Router>
Main-Office-Router>
Main-Office-Router>
Main-Office-Router>
Main-Office-Router>
Main-Office-Router>ping 10.10.42.80

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.42.80, timeout is 2 seconds:
Reply to request 0 from 10.10.42.83, 0 ms
Reply to request 0 from 10.10.42.82, 0 ms
Reply to request 1 from 10.10.42.82, 0 ms
Reply to request 1 from 10.10.42.83, 13 ms
Reply to request 2 from 10.10.42.84, 14 ms
Reply to request 2 from 10.10.42.83, 0 ms
Reply to request 2 from 10.10.42.84, 0 ms
Reply to request 3 from 10.10.42.82, 0 ms
Reply to request 3 from 10.10.42.84, 0 ms
Reply to request 4 from 10.10.42.83, 11 ms
Reply to request 4 from 10.10.42.84, 10 ms
Reply to request 4 from 10.10.42.82, 11 ms

Ctrl+F6 to exit CLI focus
```

Copy Paste

Time: 01:09:01

819HGW

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delet

Toggle PDU List Window

57°F Sunny

6:19 PM 4/18/2023

Manufacturing-PC1 (10.10.42.98)

Main-Office-Router

Physical Config CLI Attributes

IOS Command Line Interface

```
Reply to request 3 from 10.10.42.84, 0 ms
Reply to request 3 from 10.10.42.82, 0 ms
Reply to request 4 from 10.10.42.83, 11 ms
Reply to request 4 from 10.10.42.84, 10 ms
Reply to request 4 from 10.10.42.82, 11 ms

Main-Office-Router>ping 10.10.42.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.42.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Main-Office-Router>ping 10.10.42.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.42.50, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Main-Office-Router>ping 10.10.42.84

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.42.84, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Main-Office-Router>
```

Ctrl+F6 to exit CLI focus

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☐ Top

1240 PT-Router PT-Empty 1841 2620XM 26

Scenario 0 Fire Last Status Source Destination Ty